



APPENDIX E

Overcontact Binary Remnants



I only tell stories, Your Grace. They may be truths, they may be fictions. All I know
is that the stories themselves exist and that I must tell them.

— *Hoid* —



In Table E.1, we list the binaries found as the matches to the BPASS v2 models in Section 5.1.2.

Binary		Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
Name	Z	M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C	
LSS 3074	z020	30	0.3	0.6	17	1.4	0.4	NS-WD	1.57	0.42	0.78 (0.05)	0.53 (0.16)	14.27	Hobbs et al. (2005)
LSS 3074	z020	30	0.3	0.6	17	1.4	1.2	NS-NS	1.40	1.53	14.59 (0.74)	0.87 (0.90)	314.01	Verbunt et al. (2017)
LSS 3074	z020	30	0.3	0.6	17	1.4	0.8	NS-NS	1.40	1.51	6.55 (0.00)	0.69 (0.00)	473.61	Richards et al. (2023)
LSS 3074	z020	25	0.4	0.6	18	1.4	1.0	NS-NS	1.40	1.45	Unbound (38.2%)			Hobbs et al. (2005)
LSS 3074	z020	25	0.4	0.6	18	1.4	0.2	NS-WD	1.83	0.76	1.62 (0.33)	0.40 (0.31)	91.40	Verbunt et al. (2017)
LSS 3074	z020	25	0.4	0.6	18	1.4	0.6	NS-NS	1.71	1.45	4.33 (0.00)	0.03 (0.00)	1382.52	Richards et al. (2023)
LSS 3074	z020	25	0.5	0.6	25	1.4	0.6	NS-NS	1.62	1.51	4.11 (0.03)	0.87 (0.08)	9.97	Hobbs et al. (2005)
LSS 3074	z020	25	0.5	0.6	25	1.4	0.2	NS-WD	1.79	1.07	Unbound (54.0%)			Verbunt et al. (2017)
LSS 3074	z020	25	0.5	0.6	25	1.4	1.0	NS-NS	1.47	1.46	Unbound (100.0%)			Richards et al. (2023)
LSS 3074	z020	24	0.4	0.6	18	1.4	1.0	NS-NS	1.40	1.45	13.07 (0.53)	0.88 (0.11)	181.22	Hobbs et al. (2005)
LSS 3074	z020	24	0.4	0.6	18	1.4	1.0	NS-NS	1.40	1.45	10.82 (0.62)	0.65 (0.44)	2632.05	Verbunt et al. (2017)
LSS 3074	z020	24	0.4	0.6	18	1.4	0.6	NS-NS	1.71	1.45	3.99 (0.00)	0.22 (0.00)	931.16	Richards et al. (2023)
LSS 3074	z020	24	0.5	0.6	24	1.4	0.6	NS-NS	1.58	1.47	5.52 (0.19)	0.49 (0.13)	1056.95	Hobbs et al. (2005)
LSS 3074	z020	24	0.5	0.6	24	1.4	0.2	NS-WD	1.79	1.04	1.46 (0.30)	0.39 (0.27)	54.78	Verbunt et al. (2017)
LSS 3074	z020	24	0.5	0.6	24	1.4	1.0	NS-NS	1.40	1.45	14.44 (0.00)	0.61 (0.00)	7860.26	Richards et al. (2023)

Binary		Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
Name	Z	M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C	
MY Cam	z020	70	0.3	1.0	35	3.16	1.2	BH-NS	3.17	1.50	43.72 (0.78)	0.98 (0.66)	6.58	Hobbs et al. (2005)
MY Cam	z020	70	0.3	1.0	35	3.16	0.2	BH-WD	4.26	0.87	Unbound (50.2%)			Verbunt et al. (2017)
MY Cam	z020	70	0.3	1.0	35	3.16	1.4	BH-NS	3.16	1.56	Unbound (100.0%)			Richards et al. (2023)
MY Cam	z020	60	0.3	0.8	30	1.4	0.8	NS-NS	1.54	1.53	22.05 (0.73)	0.86 (0.18)	1127.07	Hobbs et al. (2005)
MY Cam	z020	60	0.3	0.8	30	1.4	1.6	NS-NS	1.40	1.51	Unbound (28.2%)			Verbunt et al. (2017)
MY Cam	z020	60	0.3	0.8	30	1.4	1.2	NS-NS	1.40	1.46	30.10 (0.00)	0.69 (0.00)	30142.89	Richards et al. (2023)
MY Cam	z020	60	0.4	0.8	40	3.16	1.0	BH-NS	3.42	1.52	27.08 (0.47)	0.92 (0.22)	167.28	Hobbs et al. (2005)
MY Cam	z020	60	0.4	0.8	40	3.16	0.2	BH-WD	4.21	0.90	Unbound (48.2%)			Verbunt et al. (2017)
MY Cam	z020	60	0.4	0.8	40	3.16	1.2	BH-NS	3.26	1.47	454.07 (0.00)	0.97 (0.00)	10326.72	Richards et al. (2023)
MY Cam	z020	60	0.3	1.0	30	3.16	1.6	BH-NS	3.17	1.51	Unbound (82.7%)			Hobbs et al. (2005)
MY Cam	z020	60	0.3	1.0	30	3.16	1.8	BH-NS	3.17	1.52	Unbound (39.9%)			Verbunt et al. (2017)
MY Cam	z020	60	0.3	1.0	30	3.16	1.4	BH-NS	3.17	1.50	68.63 (0.00)	0.52 (0.00)	434468.74	Richards et al. (2023)
MY Cam	z020	50	0.5	0.6	40	1.4	1.4	NS-NS	1.40	1.48	Unbound (73.7%)			Hobbs et al. (2005)
MY Cam	z020	50	0.5	0.6	40	1.4	0.2	NS-WD	1.73	1.26	0.94 (0.03)	0.53 (0.30)	8.31	Verbunt et al. (2017)

Binary Name		Z		Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
		M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C			
MY Cam	z020	50	0.5	0.6	40	1.4	1.0	NS-NS	1.54	1.56		Unbound (100.0%)		Richards et al. (2023)		
MY Cam	z020	50	0.4	0.8	35	1.4	0.8	NS-NS	1.59	1.51	15.94 (0.34)	0.91 (0.06)	121.62	Hobbs et al. (2005)		
MY Cam	z020	50	0.4	0.8	35	1.4	1.6	NS-NS	1.40	1.46		Unbound (40.2%)		Verbunt et al. (2017)		
MY Cam	z020	50	0.4	0.8	35	1.4	1.2	NS-NS	1.40	1.56	70.36 (0.00)	0.33 (0.00)	1745452.95	Richards et al. (2023)		
MY Cam	z020	50	0.5	0.8	40	1.4	0.8	NS-NS	1.62	1.53		Unbound (50.0%)		Hobbs et al. (2005)		
MY Cam	z020	50	0.5	0.8	40	1.4	0.2	NS-WD	1.73	1.26	1.72 (0.51)	0.48 (0.32)	51.11	Verbunt et al. (2017)		
MY Cam	z020	50	0.5	0.8	40	1.4	1.2	NS-NS	1.47	1.56	60.65 (0.00)	0.62 (0.00)	306467.37	Richards et al. (2023)		
MY Cam	z020	50	0.6	0.8	50	1.4	1.0	NS-NS	1.65	2.29		Unbound (76.0%)		Hobbs et al. (2005)		
MY Cam	z020	50	0.6	0.8	50	1.4	1.8	NS-BH	1.40	3.15		Unbound (60.4%)		Verbunt et al. (2017)		
MY Cam	z020	50	0.6	0.8	50	1.4	1.4	NS-BH	1.47	2.98	115.83 (0.00)	0.59 (0.00)	1275227.75	Richards et al. (2023)		
MY Cam	z020	40	0.7	0.8	50	1.4	1.0	NS-NS	1.65	2.29		Unbound (51.4%)		Hobbs et al. (2005)		
MY Cam	z020	40	0.7	0.8	50	1.4	0.2	NS-WD	1.76	1.98	1.08 (0.03)	0.45 (0.20)	11.60	Verbunt et al. (2017)		
MY Cam	z020	40	0.7	0.8	50	1.4	1.4	NS-BH	1.47	2.98		Unbound (100.0%)		Richards et al. (2023)		
SMC 108086*	z004	24	0.5	0.4	-	24.0	-	NS-BH	1.00	2.91		Unbound (100.0%)		Hobbs et al. (2005)		

Binary Name	Z	Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
		M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C	
SMC 108086*	z004	24	0.5	0.4	-	24.0	-	NS-BH	1.00	2.91	Unbound (100.0%)			Verbunt et al. (2017)
SMC 108086*	z004	24	0.5	0.4	-	24.0	-	NS-BH	1.00	2.91	Unbound (100.0%)			Richards et al. (2023)
SMC 108086	z004	23	0.4	0.4	18	1.4	0.4	NS-NS	1.92	1.46	7.88 (1.00)	0.58 (0.35)	1411.28	Hobbs et al. (2005)
SMC 108086	z004	23	0.4	0.4	18	1.4	1.0	NS-NS	1.61	1.44	10.63 (0.57)	0.67 (0.28)	1931.17	Verbunt et al. (2017)
SMC 108086	z004	23	0.4	0.4	18	1.4	0.6	NS-NS	1.53	1.47	6.95 (0.00)	0.43 (0.00)	2571.06	Richards et al. (2023)
SMC 108086*	z004	23	0.5	0.4	-	23.0	-	NS-BH	1.00	2.77	Unbound (100.0%)			Hobbs et al. (2005)
SMC 108086*	z004	23	0.5	0.4	-	23.0	-	NS-BH	1.00	2.77	Unbound (100.0%)			Verbunt et al. (2017)
SMC 108086*	z004	23	0.5	0.4	-	23.0	-	NS-BH	1.00	2.77	Unbound (100.0%)			Richards et al. (2023)
SMC 108086	z004	22	0.4	0.4	17	1.4	0.4	NS-NS	2.17	1.44	5.83 (1.30)	0.78 (0.48)	96.60	Hobbs et al. (2005)
SMC 108086	z004	22	0.4	0.4	17	1.4	1.0	NS-NS	1.62	1.44	13.55 (0.91)	0.93 (1.02)	31.29	Verbunt et al. (2017)
SMC 108086	z004	22	0.4	0.4	17	1.4	0.6	NS-NS	1.54	1.45	4.79 (0.00)	0.96 (0.00)	0.42	Richards et al. (2023)
SMC 108086*	z004	22	0.5	0.4	-	22.0	-	NS-BH	1.00	2.56	Unbound (100.0%)			Hobbs et al. (2005)
SMC 108086*	z004	22	0.5	0.4	-	22.0	-	NS-BH	1.00	2.56	Unbound (100.0%)			Verbunt et al. (2017)
SMC 108086*	z004	22	0.5	0.4	-	22.0	-	NS-BH	1.00	2.56	Unbound (100.0%)			Richards et al. (2023)
SMC 108086	z004	21	0.4	0.4	16	1.4	0.4	NS-NS	1.61	1.47	6.98 (0.56)	0.33 (0.30)	3368.40	Hobbs et al. (2005)

Binary Name	Z	Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
		M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C	
SMC 108086	z004	21	0.4	0.4	16	1.4	1.0	NS-NS	1.70	1.46	11.09 (0.93)	0.84 (0.91)	256.61	Verbunt et al. (2017)
SMC 108086	z004	21	0.4	0.4	16	1.4	0.6	NS-NS	1.64	1.46	3.75 (0.00)	0.58 (0.00)	226.10	Richards et al. (2023)
SMC 108086*	z004	21	0.5	0.4	-	21.0	-	NS-NS	1.00	2.40	Unbound (100.0%)			Hobbs et al. (2005)
SMC 108086*	z004	21	0.5	0.4	-	21.0	-	NS-NS	1.00	2.40	Unbound (100.0%)			Verbunt et al. (2017)
SMC 108086*	z004	21	0.5	0.4	-	21.0	-	NS-NS	1.00	2.40	Unbound (100.0%)			Richards et al. (2023)
SMC 108086	z004	20	0.4	0.4	15	1.4	1.0	NS-NS	1.53	1.46	20.76 (0.90)	0.75 (0.73)	5803.72	Hobbs et al. (2005)
SMC 108086	z004	20	0.4	0.4	15	1.4	1.0	NS-NS	1.53	1.46	Unbound (50.4%)			Verbunt et al. (2017)
SMC 108086	z004	20	0.4	0.4	15	1.4	0.6	NS-NS	1.65	1.45	5.64 (0.00)	0.64 (0.00)	442.55	Richards et al. (2023)
SMC 108086	z004	20	0.5	0.4	20	1.4	0.4	NS-NS	1.89	1.47	4.42 (0.18)	0.63 (0.10)	221.87	Hobbs et al. (2005)
SMC 108086	z004	20	0.5	0.4	20	1.4	0.2	NS-WD	1.81	0.58	1.63 (0.33)	0.39 (0.38)	121.66	Verbunt et al. (2017)
SMC 108086	z004	20	0.5	0.4	20	1.4	0.8	NS-NS	1.59	1.48	5.19 (0.00)	0.94 (0.00)	1.88	Richards et al. (2023)
SMC 108086	z004	19	0.4	0.4	15	1.4	1.0	NS-NS	1.53	1.46	17.93 (0.88)	0.66 (0.71)	8651.22	Hobbs et al. (2005)
SMC 108086	z004	19	0.4	0.4	15	1.4	0.2	NS-WD	1.85	0.34	2.85 (0.51)	0.96 (0.62)	0.36	Verbunt et al. (2017)
SMC 108086	z004	19	0.4	0.4	15	1.4	0.6	NS-NS	1.65	1.45	7.03 (0.00)	0.33 (0.00)	3429.63	Richards et al. (2023)
TU Mus	z020	25	0.3	0.4	16	1.4	0.2	NS-WD	1.60	0.95	4.32 (1.46)	0.97 (0.44)	0.12	Hobbs et al. (2005)

Binary		Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
Name	Z	M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C	
TU Mus	z020	25	0.3	0.4	16	1.4	0.2	NS-WD	1.60	0.95	0.87 (0.05)	0.41 (0.24)	15.25	Verbunt et al. (2017)
TU Mus	z020	25	0.3	0.4	16	1.4	0.6	NS-NS	1.72	1.44	4.60 (0.00)	0.16 (0.00)	1484.94	Richards et al. (2023)
TU Mus	z020	24	0.3	0.4	16	1.4	0.2	NS-WD	1.60	0.95	Unbound (89.9%)			Hobbs et al. (2005)
TU Mus	z020	24	0.3	0.4	16	1.4	0.8	NS-NS	1.40	1.44	10.93 (0.93)	0.90 (0.86)	60.41	Verbunt et al. (2017)
TU Mus	z020	24	0.3	0.4	16	1.4	0.6	NS-NS	1.72	1.44	2.73 (0.00)	0.63 (0.00)	66.09	Richards et al. (2023)
TU Mus	z020	23	0.3	0.4	15	1.4	0.8	NS-NS	1.41	1.48	8.44 (0.52)	0.45 (0.40)	4301.09	Hobbs et al. (2005)
TU Mus	z020	23	0.3	0.4	15	1.4	0.2	NS-WD	1.59	0.93	Unbound (43.6%)			Verbunt et al. (2017)
TU Mus	z020	23	0.3	0.4	15	1.4	0.6	NS-NS	1.81	1.45	3.45 (0.00)	0.12 (0.00)	685.72	Richards et al. (2023)
TU Mus	z020	16	0.6	0.4	21	1.4	0.6	NS-NS	1.64	1.50	2.76 (0.01)	0.88 (0.11)	2.73	Hobbs et al. (2005)
TU Mus	z020	16	0.6	0.4	21	1.4	1.6	NS-NS	1.40	1.50	Unbound (19.5%)			Verbunt et al. (2017)
TU Mus	z020	16	0.6	0.4	21	1.4	1.2	NS-NS	1.40	1.48	31.09 (0.00)	0.76 (0.00)	15597.63	Richards et al. (2023)
V382 Cyg	z020	50	0.2	0.6	20	1.4	0.8	NS-NS	1.40	1.44	Unbound (45.2%)			Hobbs et al. (2005)
V382 Cyg	z020	50	0.2	0.6	20	1.4	1.4	NS-NS	1.40	1.47	Unbound (51.9%)			Verbunt et al. (2017)
V382 Cyg	z020	50	0.2	0.6	20	1.4	1.0	NS-NS	1.40	1.55	12.17 (0.00)	0.77 (0.00)	1098.05	Richards et al. (2023)
V382 Cyg	z020	40	0.3	0.6	23	1.4	1.2	NS-NS	1.40	1.47	19.61 (0.60)	0.70 (0.35)	8667.25	Hobbs et al. (2005)

Binary Name	Z	Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
		M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C	
V382 Cyg	z020	40	0.3	0.6	23	1.4	0.2	NS-WD	1.80	0.99		Unbound (40.4%)		Verbunt et al. (2017)
V382 Cyg	z020	40	0.3	0.6	23	1.4	0.8	NS-NS	1.41	1.45	26.07 (0.00)	0.58 (0.00)	45411.40	Richards et al. (2023)
V382 Cyg	z020	40	0.4	0.6	30	1.4	0.6	NS-WD	1.66	0.28		Unbound (97.3%)		Hobbs et al. (2005)
V382 Cyg	z020	40	0.4	0.6	30	1.4	1.4	NS-NS	1.40	1.51		Unbound (49.9%)		Verbunt et al. (2017)
V382 Cyg	z020	40	0.4	0.6	30	1.4	1.0	NS-NS	1.40	1.45	18.06 (0.00)	0.75 (0.00)	4347.70	Richards et al. (2023)
V382 Cyg	z020	35	0.4	0.6	25	1.4	1.2	NS-NS	1.46	1.45		Unbound (32.5%)		Hobbs et al. (2005)
V382 Cyg	z020	35	0.4	0.6	25	1.4	0.2	NS-WD	1.79	1.07		Unbound (44.4%)		Verbunt et al. (2017)
V382 Cyg	z020	35	0.4	0.6	25	1.4	0.8	NS-NS	1.47	1.49	13.49 (0.00)	0.67 (0.00)	3944.11	Richards et al. (2023)
V382 Cyg	z020	30	0.4	0.6	23	1.4	1.2	NS-NS	1.40	1.47		Unbound (38.9%)		Hobbs et al. (2005)
V382 Cyg	z020	30	0.4	0.6	23	1.4	0.2	NS-WD	1.80	0.99	1.56 (0.24)	0.30 (0.35)	86.59	Verbunt et al. (2017)
V382 Cyg	z020	30	0.4	0.6	23	1.4	0.8	NS-NS	1.41	1.45	19.92 (0.00)	0.50 (0.00)	34461.76	Richards et al. (2023)
V382 Cyg	z020	30	0.5	0.6	30	1.4	0.6	NS-WD	1.66	0.28		Unbound (98.6%)		Hobbs et al. (2005)

Binary		Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
Name	Z	M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C	
V382 Cyg	z020	30	0.5	0.6	30	1.4	1.4	NS-NS	1.40	1.51	Unbound (48.3%)			Verbunt et al. (2017)
V382 Cyg	z020	30	0.5	0.6	30	1.4	1.0	NS-NS	1.40	1.45	28.16 (0.00)	0.99 (0.00)	0.88	Richards et al. (2023)
VFTS 352	z010	40	0.5	0.6	40	3.16	1.2	BH-BH	3.33	3.52	Unbound (73.3%)			Hobbs et al. (2005)
VFTS 352	z010	40	0.5	0.6	40	3.16	1.4	BH-BH	3.32	3.92	Unbound (55.4%)			Verbunt et al. (2017)
VFTS 352	z010	40	0.5	0.6	40	3.16	1.0	BH-BH	3.35	3.18	29.62 (0.00)	0.90 (0.00)	240.49	Richards et al. (2023)

Table E.1: The remnants generated by each model from the BPASS release. Post-secondary supernova properties are generated using the mentioned kick, simulating 1,000 kicks and taking the mean post-secondary parameters. Masses are measured in $M_\odot$, separations in $R_\odot$, and coalescence times in Gyr. Asterisks next to binary names denote candidates for a star that undergoes quasi-chemically homogenous evolution in the secondary. In this table, a white dwarf is left if the CO core mass of the star is less than $1.38 M_\odot$, the total mass of the star is less than $1.5 M_\odot$, or if core carbon burning does not occur. Otherwise if all three of those conditions are met, the mass boundary demarcating black holes from neutron stars is placed at $2.5 M_\odot$. Quoted errors on a_0 and e_0 , shown in parentheses underneath the nominal value, are the 68 per cent confidence intervals, and systems flagged as “Unbound” have the percentage of simulated kick events in which they are unbound displayed in parentheses, though for these systems the mean eccentricity falls outside the range of $[0, 1)$.